



Interim Report

Industry Review

1)Current practices

Banks' requirement for collaterals when making loans reduces the adverse selection problem, which in turn leads to lower default rate and also evaluates the application using their individual evaluation method to determine the credit worthiness of the company.

2)Background Research

Bank lending has been a major source of funding for most businesses. There is the need therefore for banks and other financial institutions to be careful in their loan administration to prevent the inherent risk associated with the product in order to maximize shareholders returns and enhance the image of the institution.

Lending is very risky in that repayment of the loans is not always guaranteed and most of the times depend on other factors not in the control of the borrower. Therefore, managing loans in a proper way not only has positive effect on the bank's performance but also on the borrower and a country's economy as a whole. Failure to manage loans, which make up the largest share of banks assets, would likely lead to high levels of non -performing loans. And this in turn effect on the performance of Banks and the economy at large.

- **Literature Survey**

1)Publication

Loan default can be defined as the inability of a borrower to fulfil his or her loan obligation as at when due (Balogun and Alimi, 1988). High default rates in SMEs lending should be of major concern to policy makers in developing countries, because of its unintended negative impacts on SMEs financing. Von-Pischke (1980) states that some of the impacts associated with default include: the inability to recycle funds to other borrowers; unwillingness of other financial intermediaries to serve the needs of small borrowers; and the creation of distrust.

As noted by Baku and Smith (1998), the costs of loan delinquencies would be felt by both the lenders and the borrowers. The lender has costs in delinquency situations, including lost interest, opportunity cost of principal, legal fees and related costs. For the borrower, the decision to default is a trade-off between the penalties in lost reputation from default versus the opportunity cost of forgoing investments due to working out the current loan.

2)Undergoing research

The research design for the current study refers to a quantitative form. This research concentrates on the relationship among variables more than on testing activity impact, and uses correlation design. Based on the described research objective, this study will adopt a correlation design. Correlation design allows us to predict an outcome and know the relation between variables.

Dataset and Domain

- Data Dictionary

Variable Name	Description
UniqueID	Identifier for customers
loan_default	Payment default in the first EMI on due date
disbursed_amount	Amount of Loan disbursed
asset_cost	Cost of the Asset
ltv	Loan to Value of the asset
branch_id	Branch where the loan was disbursed
supplier_id	Vehicle Dealer where the loan was disbursed
manufacturer_id	Vehicle manufacturer(Hero, Honda, TVS etc.)
Current_pincode	Current pincode of the customer
Date.of.Birth	Date of birth of the customer
Employment.Type	Employment Type of the customer (Salaried/Self Employed)
DisbursalDate	Date of disbursement
State_ID	State of disbursement
Employee_code_ID	Employee of the organization who logged the disbursement
MobileNo_Avl_Flag	if Mobile no. was shared by the customer then flagged as 1
Aadhar_flag	if aadhar was shared by the customer then flagged as 1
PAN_flag	if pan was shared by the customer then flagged as 1
VoterID_flag	if voter was shared by the customer then flagged as 1
Driving_flag	if DL was shared by the customer then flagged as 1
Passport_flag	if passport was shared by the customer then flagged as 1
PERFORM_CNS.SCORE	Bureau Score
PERFORM_CNS.SCORE.DESCRPTION	Bureau score description
PRI.NO.OF.ACCTS	count of total loans taken by the customer at the time of disbursement
PRI.ACTIVE.ACCTS	count of active loans taken by the customer at the time of disbursement
PRI.OVERDUE.ACCTS	count of default accounts at the time of disbursement
PRI.CURRENT.BALANCE	total Principal outstanding amount of the active loans at the time of disbursement
PRI.SANCTIONED.AMOUNT	total amount that was sanctioned for all the loans at the time of disbursement
PRI.DISBURSED.AMOUNT	total amount that was disbursed for all the loans at the time of disbursement
SEC.NO.OF.ACCTS	count of total loans taken by the customer at the time of disbursement
SEC.ACTIVE.ACCTS	count of active loans taken by the customer at the time of disbursement
SEC.OVERDUE.ACCTS	count of default accounts at the time of disbursement
SEC.CURRENT.BALANCE	total Principal outstanding amount of the active loans at the time of disbursement
SEC.SANCTIONED.AMOUNT	total amount that was sanctioned for all the loans at the time of disbursement
SEC.DISBURSED.AMOUNT	total amount that was disbursed for all the loans at the time of disbursement
PRIMARY.INSTAL.AMT	EMI Amount of the primary loan
SEC.INSTAL.AMT	EMI Amount of the secondary loan
NEW.ACCTS.IN.LAST.SIX.MONTHS	New loans taken by the customer in last 6 months before the disbursment

DELINQUENT.ACCTS.IN.LAST.SIX.MONTHS	Loans defaulted in the last 6 months
AVERAGE.ACCT.AGE	Average loan tenure
CREDIT.HISTORY.LENGTH	Time since first loan
NO.OF_INQUIRIES	Enquiries done by the customer for loans

•Variable Categorization (count of numeric and categorical)

A variable can be defined as something that is subject to change. There are two types of variables, namely:

- Numerical
- Categorical

Numerical data refers to the data that is in the form of numbers, and not in any language or descriptive form.

The number of numerical features are: 35

The numerical features are:

'UNIQUEID', 'DISBURSED_AMOUNT', 'ASSET_COST', 'LTV', 'BRANCH_ID', 'SUPPLIER_ID', 'MANUFACTURER_ID', 'CURRENT_PINCODE_ID', 'STATE_ID', 'EMPLOYEE_CODE_ID', 'MOBILE_NO_AVL_FLAG', 'AADHAR_FLAG', 'PAN_FLAG', 'VOTERID_FLAG', 'DRIVING_FLAG', 'PASSPORT_FLAG', 'PERFORM_CNS_SCORE', 'PRI_NO_OF_ACCTS', 'PRI_ACTIVE_ACCTS', 'PRI_OVERDUE_ACCTS', 'PRI_CURRENT_BALANCE', 'PRI_SANCTIONED_AMOUNT', 'PRI_DISBURSED_AMOUNT', 'SEC_NO_OF_ACCTS', 'SEC_ACTIVE_ACCTS', 'SEC_OVERDUE_ACCTS', 'SEC_CURRENT_BALANCE', 'SEC_SANCTIONED_AMOUNT', 'SEC_DISBURSED_AMOUNT', 'PRIMARY_INSTALL_AMT', 'SEC_INSTALL_AMT', 'NEW_ACCTS_IN_LAST_SIX_MONTHS', 'DELINQUENT_ACCTS_IN_LAST_SIX_MONTHS', 'NO_OF_INQUIRIES', 'LOAN_DEFAULT'.

THE NUMBER OF CATEGORICAL FEATURES IS: 6

THE CATEGORICAL FEATURES ARE:

'DATE_OF_BIRTH', 'EMPLOYMENT_TYPE', 'DISBURSAL_DATE',
'PERFORM_CNS_SCORE_DESCRIPTION', 'AVERAGE_ACCT_AGE',
'CREDIT_HISTORY_LENGTH'.

```
df_num=train.select_dtypes(include=np.number)  
len(df_num.columns)
```

35

```
df_cat=train.select_dtypes(include=object)  
len(df_cat.columns)
```

6

Pre Processing Data Analysis (count of missing/ null values, redundant columns, etc.)

Pre Processing Data Analysis:

Data Pre-processing can refer to manipulation or dropping data before it is used in order to ensure or enhance performance. It is an important step in the Data Mining Process.

1) Count of missing/null values

```
Total = train.isnull().sum().sort_values(ascending = False)
Percent = (train.isnull().sum()*100/train.isnull().count()).sort_values(ascending = False)
missing_data = pd.concat([Total, Percent], axis = 1, keys = ['Total', 'Percentage of Missing Values'])
missing_data
```

	Total	Percentage of Missing Values
EMPLOYMENT_TYPE	7661	3.285811
UNIQUEID	0	0.000000
SEC_SANCTIONED_AMOUNT	0	0.000000
PRI_OVERDUE_ACCTS	0	0.000000

Missing Value Analysis:

Missing Values are found in feature EMPLOYMENT_TYPE.

As the percentage of missing values is less than 5% in the feature, the FILLNA approach is followed to overcome the issue of missing values in the dataset.

since we didn't have enough information to fill the null values, we created a separate category unknown which represents that there is no information about the employment type for such candidates

```
train['EMPLOYMENT_TYPE']=train['EMPLOYMENT_TYPE'].fillna('Unknown')
```

```
train.EMPLOYMENT_TYPE.value_counts()
```

```
Self employed    127635
Salaried          97858
Unknown           7661
Name: EMPLOYMENT_TYPE, dtype: int64
```

2) Redundant columns:

Some columns that have totally unique values / those which are not required for the analysis are considered redundant and are dropped. Below listed are some columns which were dropped from the dataset.

'UNIQUEID','MOBILENO_AVL_FLAG','AADHAR_FLAG','PAN_FLAG','V
OTERID_FLAG','DRIVING_FLAG','PASSPORT_FLAG','SUPPLIER_ID','M
ANUFACTURER_ID','BRANCH_ID','CURRENT_PINCODE_ID'.

•Project Justification -

1.Project statement

Project is based on predicting the defaulting loans. The predictions are done by evaluating various factors that influence the repayment capacity of the loan applicant. The project aims to create a machine learning model which is highly accurate in predicting such defaults.

2. outcome

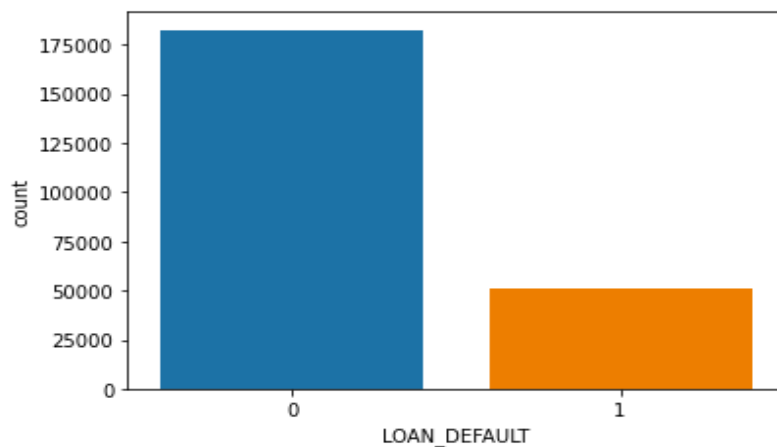
The costs of loan delinquencies would be felt by both the lenders and the borrowers. The lender has costs in delinquency situations, including lost interest, opportunity cost of principal, legal fees and related costs. For the borrower, the decision to default is a trade-off between the penalties in lost reputation from default versus the opportunity cost of forgoing investments due to working out the current loan.

Data Exploration (EDA)

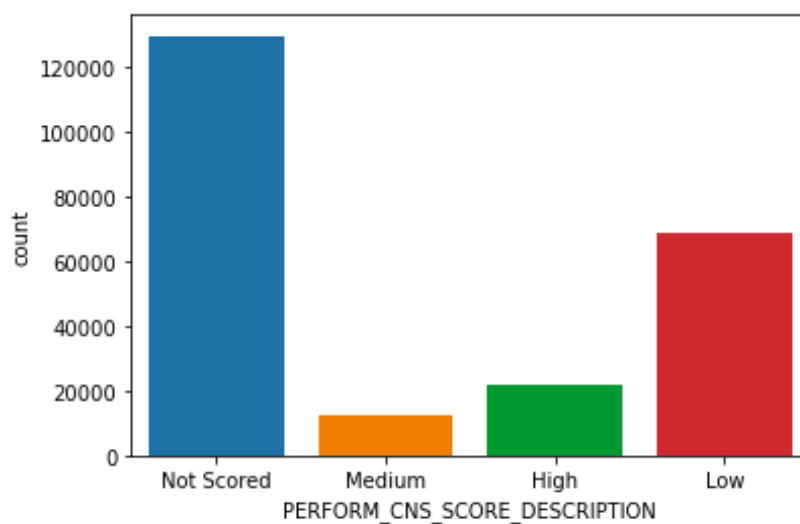
Relationship between variables:

1) Univariate Analysis:

Univariate data visualization plots help us comprehend the enumerative properties as well as a descriptive summary of the particular data variable. These plots help in understanding the location/position of observations in the data variable, its distribution, and dispersion.



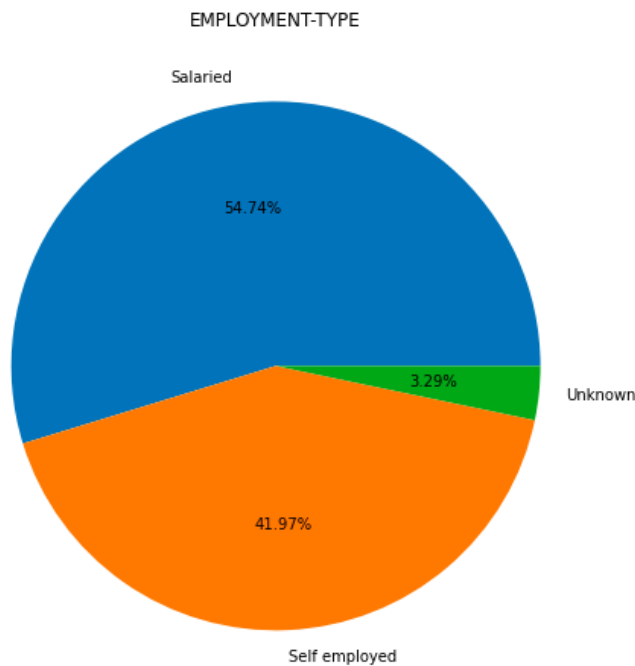
Interpretation: The target variable seems to be imbalanced so we will need to use SMOTE analysis at the time model building.



Interpretation:

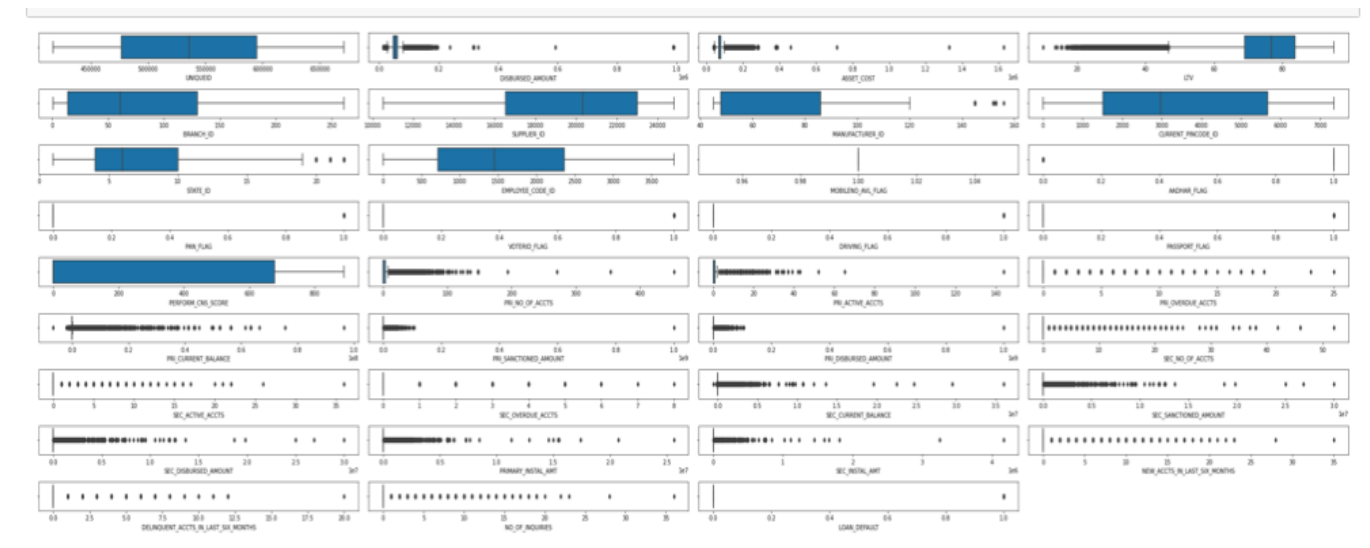
On visualizing the categorical columns, we can figure that:

- 1) There are high number of customers who do not have a bureau history record.
- 2) There are around 20000 customers who are at high risk because of their low bureau score.



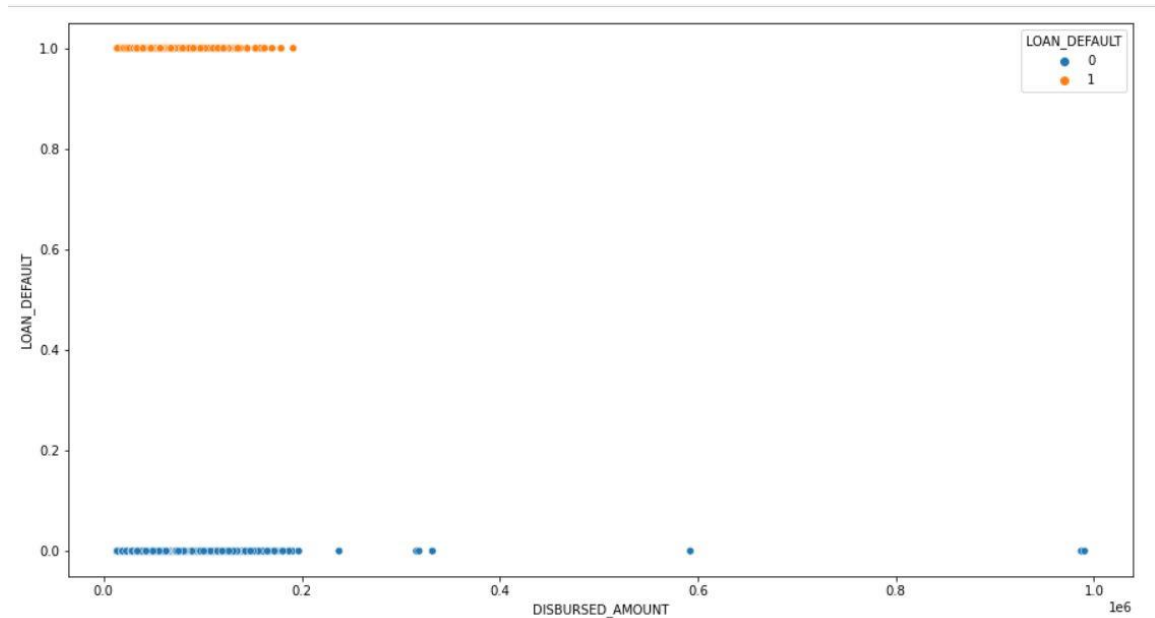
Interpretation: On visualizing the categorical columns, we can figure that:

The count of Self-employed customers are more than the employed customers in the data.

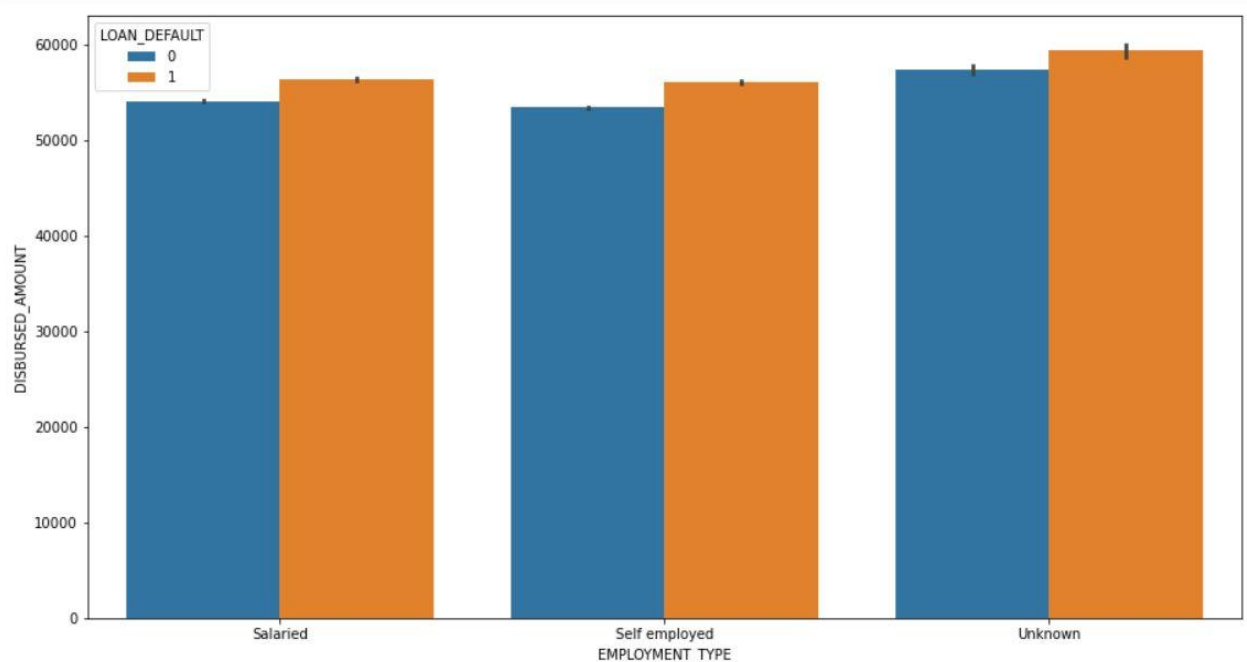


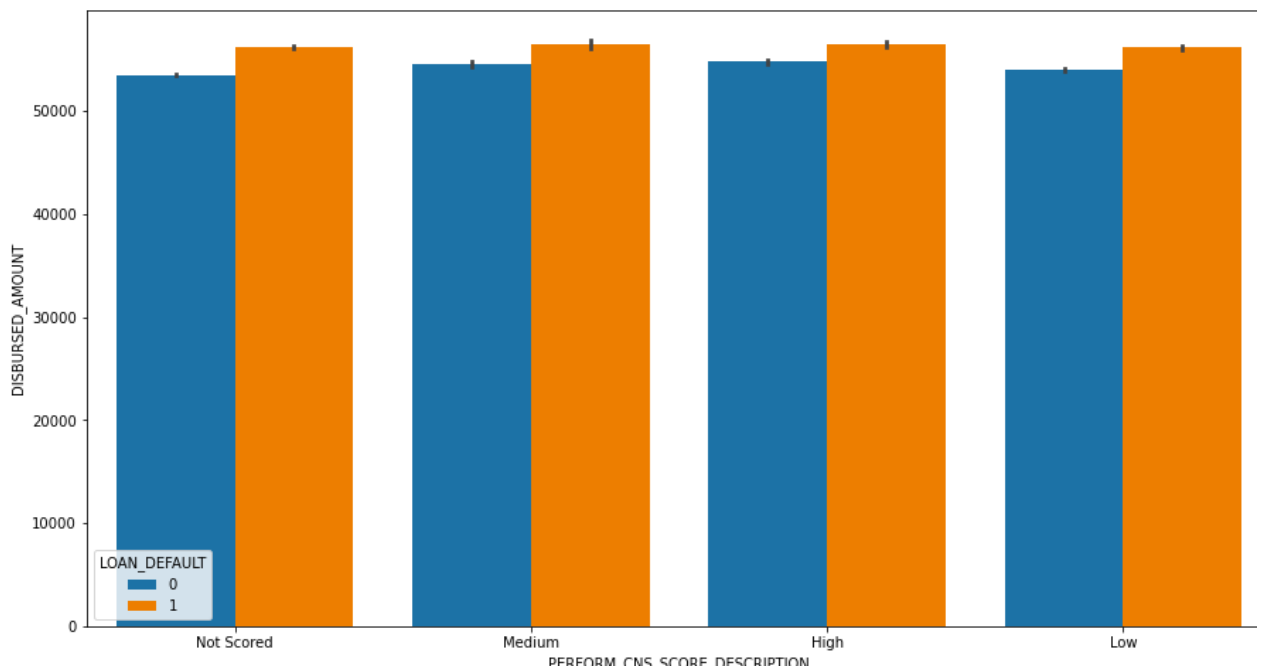
2) **Bivariate Analysis:** Bivariate analysis is one of the statistical analyses where two variables are observed. One variable here is dependent while the other is independent.

These variables are usually denoted by X and Y. So, here we analyze the changes that occurred between the two variables and to what extent.



3) **Multivariate Analysis:** Multivariate analysis is based on the statistical principle of multivariate statistics, which involves observation and analysis of more than one statistical outcome variable at a time.

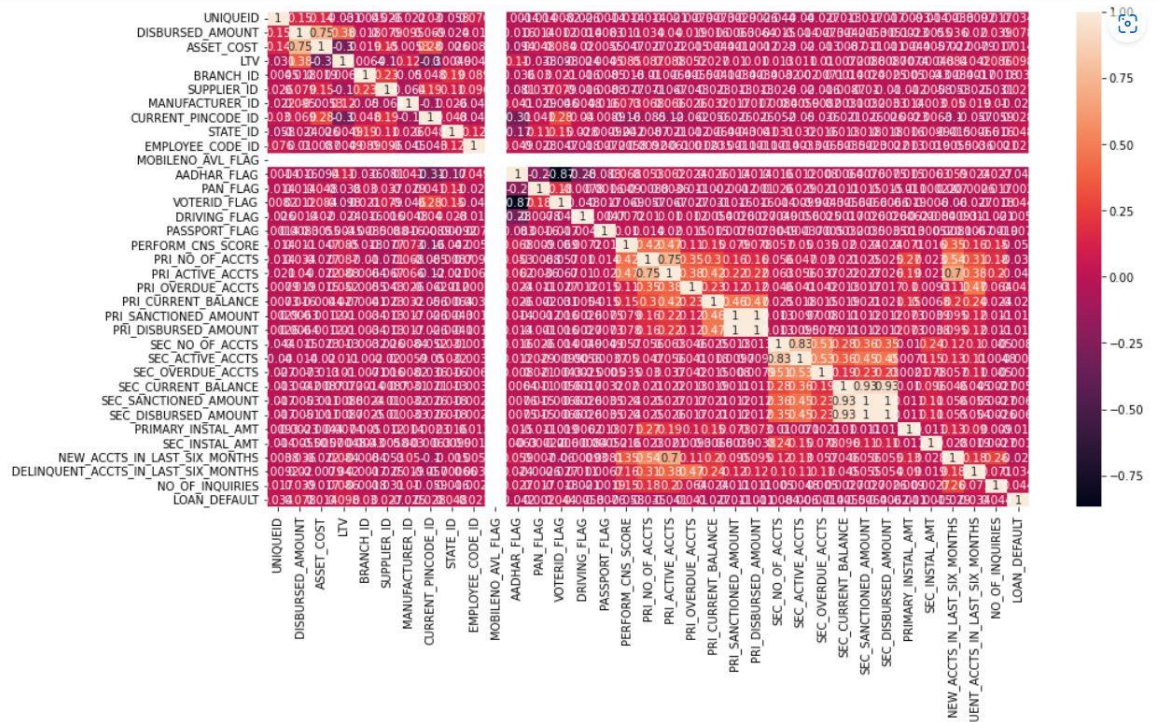




Check for multi-collinearity

Multicollinearity:

Multicollinearity occurs when two or more independent variables(also known as predictors) are highly correlated with one another in a regression model.This means that an independent variable can be predicted from another independent variable in a regression model. In this dataset, there is only one numerical feature and so there is no scope for multicollinearity.



Disbursed amount and asset cost highly correlated, also correlated to ltv

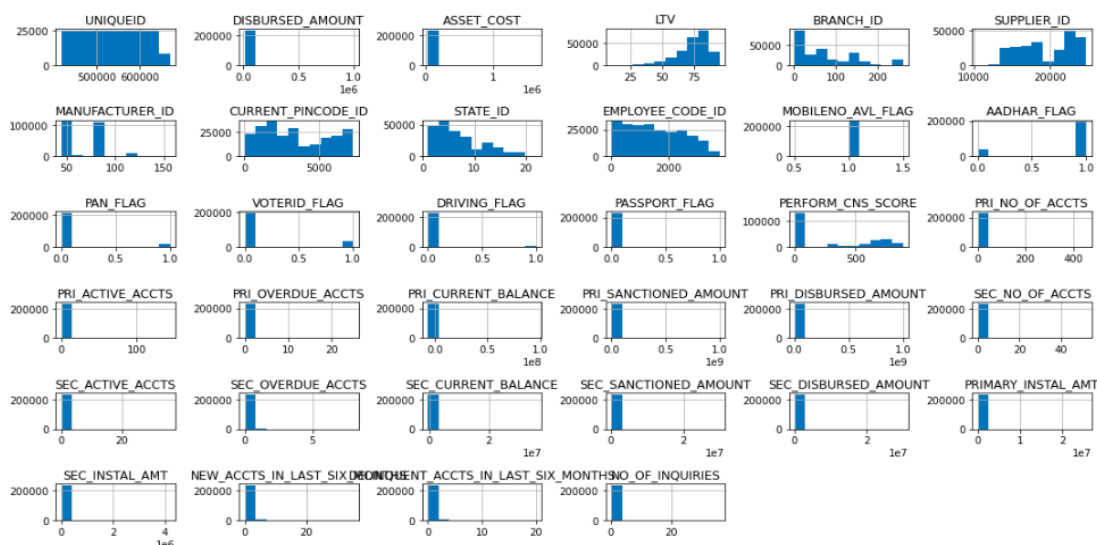
Average account age and credit history length correlated

Predictors SEC also highly correlated

PRI. OVERDUE.ACCTS and

DELINQUENT.ACCTS.IN.LAST.SIX.MONTHS highly correlated

•distribution of variables



The feature engineered Age_During_Disbursal variable is moderately skewed
The Loan to value variable is highly negatively skewed.

- **presence of outliers and its treatment**

Outliers are data points that are far from other data points. Outliers are problematic for many statistical analyses because they can cause tests to either miss significant findings or distort real results.

The features in this dataset contains outliers but it cannot be treated. According the domain knowledge removing of outliers is insignificant for the dataset.

Statistical Significance of Variables:

Surprisingly, determining which variable is the most important is more complicated than it first appears.

Statistical significance tests are designed to address this problem and quantify the likelihood of the samples of skill scores being observed given the assumption that they were drawn from the same distribution.

Changes in the independent variable are associated with changes in the dependent variable at the population level. This variable is statistically significant and probably a worthwhile addition to the model.

```

: signif_num=[]
for i in df_num:
    zero=df1.loc[df1.LOAN_DEFAULT==0,i]
    one=df1.loc[df1.LOAN_DEFAULT==1,i]
    teststats, pvalue= stats.ttest_ind(zero,one)

    if(pvalue<0.05):
        print(i)
        signif_num.append(i)

```

```

DISBURSED_AMOUNT
ASSET_COST
LTV
EMPLOYEE_CODE_ID
PERFORM_CNS_SCORE
PRI_NO_OF_ACCTS
PRI_ACTIVE_ACCTS
PRI_OVERDUE_ACCTS
PRI_CURRENT_BALANCE
PRI_SANCTIONED_AMOUNT
PRI_DISBURSED_AMOUNT
SEC_NO_OF_ACCTS
SEC_ACTIVE_ACCTS
SEC_CURRENT_BALANCE
SEC_SANCTIONED_AMOUNT
SEC_DISBURSED_AMOUNT
PRIMARY_INSTALL_AMT
NEW_ACCTS_IN_LAST_SIX_MONTHS
DELINQUENT_ACCTS_IN_LAST_SIX_MONTHS
AVERAGE_ACCT_AGE
CREDIT_HISTORY_LENGTH
NO_OF_INQUIRIES
LOAN_DEFAULT
Age

```

```

: feats=[]
for i in df_cat:
    tbl=pd.crosstab(train.loc[:,i],train.LOAN_DEFAULT)
    test_stats, pval, df, exp_freq= stats.chi2_contingency(tbl)
    if pval<0.05:
        feats.append(i)
print(feats)

```

```
['EMPLOYMENT_TYPE', 'DISBURSAL_DATE', 'PERFORM_CNS_SCORE_DESCRIPTION', 'AVERAGE_ACCT_AGE', 'CREDIT_HISTORY_LENGTH']
```

If the p-value for a variable is less than the significance level, the sample data provide enough evidence to reject the null hypothesis for the entire population. Data will favor the hypothesis that there is a non-zero correlation.

•class imbalance and its treatment



Interpretation: The target variable seems to be imbalanced so we will need to use SMOTE analysis at the time model building.

```
]: from imblearn.over_sampling import SMOTE
sm=SMOTE(random_state=100)
x_res,y_res=sm.fit_resample(X_train,Y_train)
```

Feature Engineering

Whether any transformations required

```
train = train.replace({'PERFORM_CNS_SCORE_DESCRIPTION': {'C-Very Low Risk': 'Low', 'A-Very Low Risk': 'Low',
'B-Very Low Risk': 'Low', 'D-Very Low Risk': 'Low',
'F-Low Risk': 'Low', 'E-Low Risk': 'Low', 'G-Low Risk': 'Low',
'H-Medium Risk': 'Medium', 'I-Medium Risk': 'Medium',
'J-High Risk': 'High', 'K-High Risk': 'High', 'L-Very High Risk': 'High',
'M-Very High Risk': 'High', 'Not Scored: More than 50 active Ac': 'Not Scored',
'Not Scored: Only a Guarantor': 'Not Scored', 'Not Scored: Not': 'Not Scored',
'Not Scored: No Activity seen on the customer (Inactive)': 'Not Scored',
'Not Scored: Sufficient History Not Available': 'Not Scored',
}}})
```

```
train.PERFORM_CNS_SCORE_DESCRIPTION.value_counts()
```

```
Not Scored    129785
Low           69022
High          21935
Medium        12412
Name: PERFORM_CNS_SCORE_DESCRIPTION, dtype: int64
```

The column 'Performance CNS Score Description' is been divided into various categories. we binned it into 4 categorise which are 'low risk', 'medium risk', 'high risk', ' no score'.

```
train['DATE_OF_BIRTH'] = pd.to_datetime(train['DATE_OF_BIRTH'])
train['DISBURSAL_DATE'] = pd.to_datetime(train['DISBURSAL_DATE'])
```

```
train['Age']=((train['DISBURSAL_DATE']-train['DATE_OF_BIRTH'])/365)
```

```
train['Age']=(train.DISBURSAL_DATE-train.DATE_OF_BIRTH).apply(lambda x: str(x).split(' ')[0]).apply(lambda x: int(x))
```

```
train['Age']=train['Age'].astype(int)
```

we created a new column age by subtracting the disbursal amount with date of birth which represents the age of the loan applicant when the loan was disbursed.

```
l = []
for items in train['AVERAGE_ACCT_AGE'].values:
    y = int(items[0]) * 12
    if items[6]== 'm':
        m = int(items[5])
    else:
        m = int(items[5:7])
    total = y + m
    l.append(total)

train.loc[:, 'AVERAGE_ACCT_AGE'] = l
```

```
l = []
for items in train['CREDIT_HISTORY_LENGTH'].values:
    y = int(items[0]) * 12
    if items[6]== 'm':
        m = int(items[5])
    else:
        m = int(items[5:7])
    total = y + m
    l.append(total)

train.loc[:, 'CREDIT_HISTORY_LENGTH'] = l
```

Here the 'Average Account Age' and 'Credit History Length' column are given in years and months format.

Both of the columns are further converted in months format.

Scaling the data

In Data Processing, we try to change the data in such a way that the model can process it without any problems. And Feature Scaling is one such process in which we transform the data into a better version. Feature Scaling is done to normalize the features in the dataset into a finite range.

```
from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
scaler.fit(x_res)
x_res_scaled=scaler.transform(x_res)
```

Feature selection

Final features are given by VIF. This method needs one machine learning algorithm and uses its performance as evaluation criteria. Feed the features to the selected Machine Learning algorithm and based on the model performance we add/remove the features.

```
for ind in range(len(df_numeric_features_vif.columns)):
    vif = pd.DataFrame()
    vif["VIF_Factor"] = [variance_inflation_factor(df_numeric_features_vif.values, i) for i in range(df_numeric_features_vif.columns)]
    vif["Features"] = df_numeric_features_vif.columns
    multi = vif[vif['VIF_Factor'] > 5]
    if(multi.empty == False):
        df_sorted = multi.sort_values(by = 'VIF_Factor', ascending = False)
    else:
        print(vif)
        break
    if (df_sorted.empty == False):
        df_numeric_features_vif = df_numeric_features_vif.drop(df_sorted.Features.iloc[0], axis=1)
    else:
        print(vif)
```

	VIF_Factor	Features
0	2.227572	EMPLOYMENT_TYPE_Self employed
1	2.505858	PERFORM_CNS_SCORE_DESCRIPTION_Low
2	1.291012	PERFORM_CNS_SCORE_DESCRIPTION_Medium
3	2.805374	PERFORM_CNS_SCORE_DESCRIPTION_Not Scored
4	3.264407	EMPLOYEE_CODE_ID
5	2.464870	PRI_NO_OF_ACCTS
6	1.826168	PRI_OVERDUE_ACCTS
7	1.469217	PRI_CURRENT_BALANCE
8	1.287264	PRI_SANCTIONED_AMOUNT
9	3.461626	SEC_NO_OF_ACCTS
10	3.622021	SEC_ACTIVE_ACCTS
11	1.433269	SEC_OVERDUE_ACCTS
12	1.156488	SEC_CURRENT_BALANCE
13	1.098163	PRIMARY_INSTAL_AMT
14	1.071288	SEC_INSTAL_AMT
15	1.868749	NEW_ACCTS_IN_LAST_SIX_MONTHS
16	1.482509	DELINQUENT_ACCTS_IN_LAST_SIX_MONTHS
17	4.045749	AVERAGE_ACCT_AGE
18	4.627979	CREDIT_HISTORY_LENGTH
19	1.175471	NO_OF_INQUIRIES

Now, we have all the variables with VIF less than 5. So, we can conclude that there is no multicollinearity in the data.

We will use the above dataframe "df_numeric_features_vif" to build the model with significant variables.

Assumptions

•Regression –Logistic Regression

Assumption 1: Appropriate dependent variable structure

Assumption 2: There is a linear relationship between the logit of the outcome and each independent variable.

Assumption 3: No Multicollinearity

Assumption 4: No Influential Outliers

Assumption 5: Observation Independence

• Classification –

Decision Tree

At the beginning, the whole training set is considered as the root.

Feature values are preferred to be categorical. If the values are continuous then they are discretized prior to building the model.

Records are distributed recursively on the basis of attribute values.

Order to placing attributes as root or internal node of the tree is done by using some statistical approach.

Random Forest

There should be some actual values in the feature variables of the dataset, which will give the classifier a better chance to predict accurate results, rather than provide an estimation. Missing values should be handled from training the model. The predictions from each tree must have very low correlations

----- **Interim Presentation Checkpoint** -----
